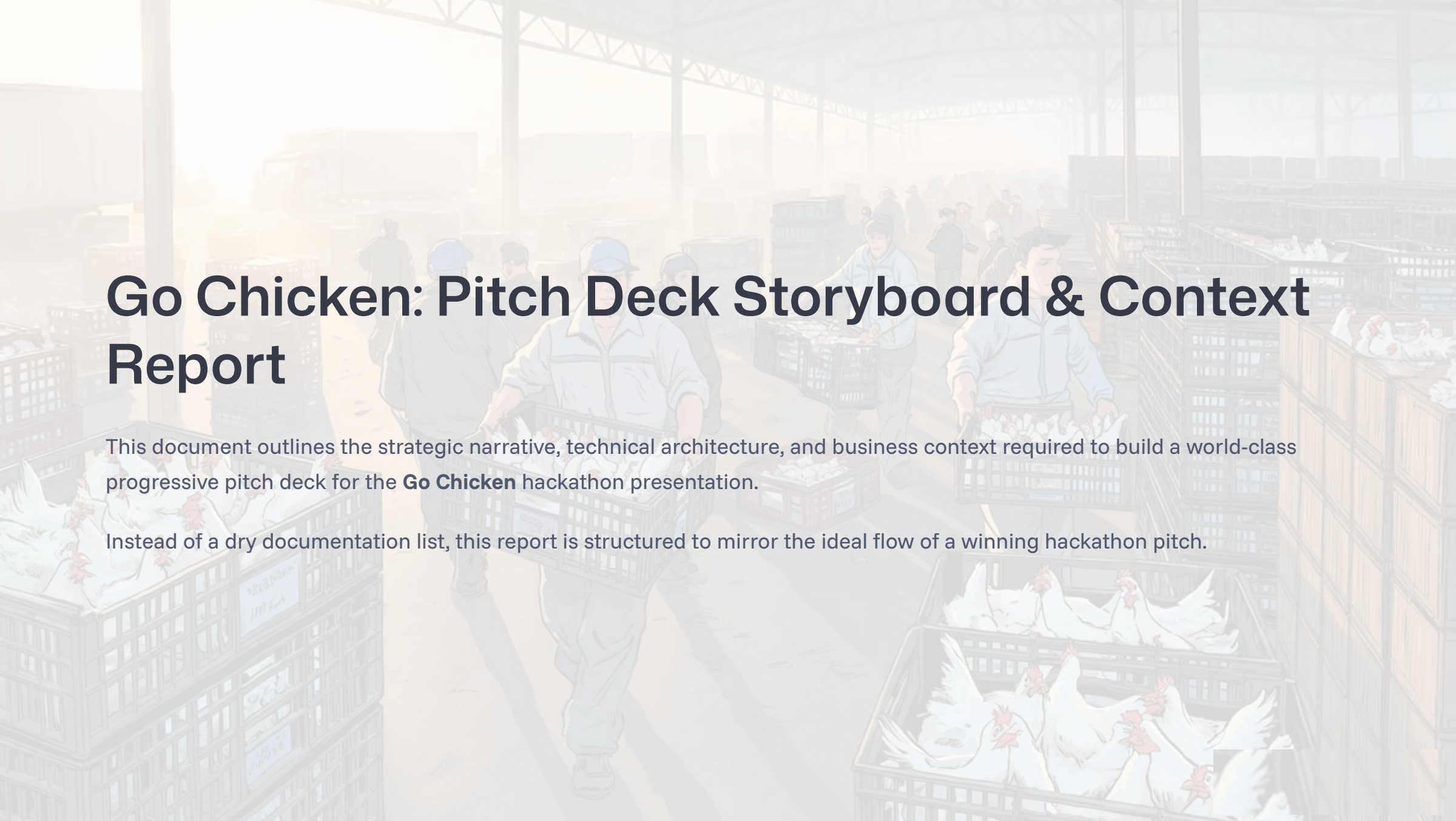




Go Chicken: Pitch Deck Storyboard & Context Report

This document outlines the strategic narrative, technical architecture, and business context required to build a world-class progressive pitch deck for the **Go Chicken** hackathon presentation.

Instead of a dry documentation list, this report is structured to mirror the ideal flow of a winning hackathon pitch.

1. The Chaos (The Hook)

The Reality of Wholesale Poultry Today

Managing wholesale poultry supply chains in emerging markets relies heavily on manual, unstructured communication (WhatsApp) and physical record-keeping (Khata books). This creates massive operational bottlenecks, pricing disputes, and lost revenue.

Latency

It takes up to 30 minutes to negotiate a rate, verify inventory, and confirm an order.

Errors

Misread WhatsApps during the morning rush lead to duplicate orders and misrouted trucks.

Pricing Disputes

Volatile live-bird prices change hourly, leading to massive margin loss when clerks quote outdated prices.

2. Meet Raj (The Persona)

To understand the problem, we need to meet the people living it.

Raj, The Poultry Wholesaler (Main Boss)

Goal: Maximize profitability and scale operations to handle more daily volume.

Reality: Deeply stressed. He is blind to real-time inventory, unaware of his exact outstanding Khata debt, and completely reliant on exhausted clerks to manually process hundreds of chaotic WhatsApp messages.

Other Stakeholders Impacted

- **Retailer (Butcher/Restaurant):** Wasting time negotiating on WhatsApp, unaware of their exact Khata debt until a dispute arises at the end of the month.
- **Delivery Driver:** Dispatched verbally, leading to miscommunications on delivery amounts and cash collection targets.

3. "What if WhatsApp could run the entire business?"

We asked ourselves: What if we didn't force retailers to download a clunky, generic ERP app? What if we brought enterprise software to where the business already happens—WhatsApp?

Our vision for Go Chicken: **Retailer sends a text → AI understands it → Backend prices it → Dashboard updates live → Ledger is reconciled.**

All under 1 second. Zero friction.

4. The Live Demo (The "Wow" Moment)

The ideal 3-minute live hackathon demo:

1

The Retailer Phone Screen (Left side of screen)

- Presenter opens WhatsApp and texts naturally: *"Need 50kg chicken, what's today's rate?"*
- The bot replies instantly (< 1s) with a dynamically generated Quote, their exact Khata balance, and interactive "Confirm / Cancel" buttons.

2

The Wholesaler Dashboard (Right side of screen)

- The presenter clicks "Confirm" on the phone.
- The audience watches the monochromatic B&W Next.js Enterprise Dashboard update *instantly* via **Server-Sent Events (SSE)** with zero page refresh. The new order appears, the inventory drops, and the Khata analytics dynamically adjust.

3

The Pricing Magic

- On the dashboard, Raj updates the price of live chicken from ₹150 to ₹160.
- Presenter sends another WhatsApp: *"Rate?"* and the AI instantly quotes the newly updated price.

5. "How did we build it?" (Technical Architecture)

We built a highly scalable, event-driven enterprise engine disguised as a chat bot.

The Data Pipeline



Frontend

Next.js (B&W Enterprise UI, SSE live updates, Optimistic Rollbacks).

Backend

FastAPI (Python), serving as both the RESTful API and the direct WhatsApp Webhook handler for sub-second latency.

Database

PostgreSQL with pgvector hosted on Supabase.

AI Processing

- *Hackathon Deployment:* **Groq** for lightning-fast intent extraction.
- *Local Development:* **Ollama** LLM.
- *Resiliency:* Fast-Fail Health Checks drop to a 100% Regex Fallback instantly if the AI is unreachable, guaranteeing uptime.

6. Enterprise Engineering Highlights

Judges love more than just a standard CRUD app. Go Chicken differentiates itself with advanced software engineering patterns:

Hierarchical Pricing Engine

Dynamically calculates pricing based on customer tiers, generating **Immutable Quote Snapshots**.

Quote-to-Order Conversion

A deterministic flow from Quote Approval → Order State Machine (no AI hallucinations allowed).

CQRS & Analytics

Read and Write models are segregated. We built an **AnalyticsService** with a **Projection Rebuilder** to serve lightning-fast dashboard metrics without locking the transactional database.

Event-Driven Architecture

Uses a **Transactional Outbox** pattern to guarantee message delivery to external systems and maintain eventual consistency.

Immutable Khata Ledger

Financial transactions are treated as append-only events, ensuring perfect financial auditability.

Server-Sent Events (SSE)

We graduated from basic polling to SSE for a truly reactive, real-time SPA experience.

Multi-Tenant Design

Strict server-side isolation prevents horizontal privilege escalation.

7. Business Value



Zero-Latency Ordering

Order processing time drops from 30 minutes to 1 second.



Margin Protection

Dynamic Pricing API and Immutable Quotes prevent clerks from accidentally selling at outdated, unprofitable rates.



Zero Khata Errors

Automated ledger updates prevent disputes and bad debt accumulation.



Seamless Adoption

Retailers don't download an app; they just use the WhatsApp they already know.

8. The Vision (Roadmap)

